

1 Provision & connect
POST /daq · test-driver

2 Bind node → line / device
PUT /daq/{id}/bindings

3 Compose & deploy team
POST /teams/{id}/deploy

4 Governed closed loop
dcw_control → dcw_judge

1 Provision & connect

one edge runtime per node

1000ms Global 1000ms mock Cast-film line A

1000ms Global 1000ms modbus-tcp Cast-film line A

Each device becomes one **node**: cadence, **driver** and line come from its export, not code.

live Cast-film line A: 4 nodes, modbus-tcp + mock, 1 s cadence.

3 Compose & deploy the agent team

one lead, harness-backed workers

Cast-film control team

Public ☒ admin ...

lead Cast-film process lead omp ×

worker DAQ engineer omp ×

worker Model trainer omp ×

A team is a reusable **role** / **harness** assignment, deployed into a channel.

ready one lead + two workers on harness omp; a mock team enables replay.

2 Bind the node to the plant

device + recipe window travel with the node

SETPOINT	PROCESS RANGE	HOLD INTERVAL	BOUND DEVICE
150~200	Apply	150 ~ 200 °C	5000ms · 5000ms
			Die-head heater

Binding gives a control node a **device twin** and a **recipe window**—node properties, so configuration re-specifies a deployment.

bound 150–200 °C window, 5 s keep-alive, device Die-head heater.

4 Governed closed loop

write → observe → judge → journal

Optimization records

judged-keep 183 → 178 keep(agent)

superseded 182 → 183 uncertain(system)

A dcw_control write opens an **optimization record**; dcw_judge resolves it to keep or rollback with a reason.

judged kept: 183 → 178 keep(agent); the superseded record carries the system verdict.